

Research Interests

Distributed Systems, Operating Systems, Concurrency, Persistent Memory, Language Runtimes, Cloud Infrastructures, Cache Coherence, Coherent Fabrics (CXL)

Education

- 2018 - 2023 **Ph.D. in Computer Science**, *Institut Polytechnique de Paris*, Palaiseau (91), France.
Thesis: “A Support for Persistent Memory in Java” – Advisors: *Pierre Sutra, Gaël Thomas*.
- 2015 - 2018 **Dipl. Ing. (Msc), in Computer Science**, *Télécom SudParis*, Évry (91), France.
Majors: *Distributed systems and computing* – Graduated first in class.
- 2013 - 2015 **MPSI/MP* (CPGE)**, *Lycée Camille Guérin* (86), France.
Intense & highly « competitive exam preparatory courses », for admission to french higher-education national engineering schools. Majors: *Mathematics, Physics*; Minor: *Computer Science*
- 2010 - 2013 **Science Baccalauréat**, High-School: *Lycée Pilote Innovant et International* (86), France.
With honours, European class in engineering sciences, Majors: *Mathematics, Engineering sciences*

Experience

Research

- Oct 2023 **Humboldt Post-doctoral Researcher**, *Technical University of Munich*, Germany.
- Present Group: *Chair of Computer Systems* – Host: *Pramod Bhatotia*
- **CXL memory consistency**: shared remote CXL memory across heterogeneous architectures.
 - **CXL bridges**: integrating CXL 3.0 cache coherence into arbitrary host architectures.
 - **CXL GPUs**: GPUs with CXL 3.0 integration for symmetric cache coherence with host CPUs.
- Oct 2018 **Graduate Research Assistant**, *Institut Polytechnique de Paris*, France.
- Feb 2023 Group: *Parallel and Distributed Systems* – Advisors: *Pierre Sutra, Gaël Thomas*
- **J-NVM**: a pure-java library to efficiently support persistent memory in Java.
 - **NVMM & FaaS**: persistent memory for stateful serverless workloads.
- Mar 2018 **Research Intern**, *IMDEA Software*, Madrid, Spain.
- Aug 2018 Advisor: *Alexey Gotsman*
- **White-box atomic multicast**: a novel & efficient distributed protocol for genuine atomic multicast. Implemented in *C* with *libevent*, evaluated in *local* and *geo-wide* area networks.

Undergraduate Research Projects

- Fall 2017 **Scalevisor**, a research microkernel and bare-metal hypervisor for NUMA machines.
Master 2 project at *Télécom SudParis*, 2-student groupwork – Advisor: *Gaël Thomas*.
- Port of the microkernel from x86 AMD platform to Intel's. (*C, x86 assembly*)
 - Add support for vendor-specific virt. extensions (*Intel VMX*), replacing *AMD-V* & *AMD-Vi*.
- Spring 2017 **Mesosearch**, investigate dynamic service scaling for orchestrated containers.
Master 1 project at *Télécom SudParis*, 4-student groupwork – Advisors: *Pierre Sutra, Gaël Thomas*.
- Deployment of a real-world distributed test app – a fully scalable and elastic web search engine using *Apache Mesos, Zookeeper, HDFS, Hadoop, Myriad, Nutch* and *Elasticsearch*.
 - Implementation of an application-oriented resource monitoring system for Mesos, providing necessary data to enable automatic and dynamic scaling.

Student Volunteering

- Oct 2015 **Sys & Net Admin at MiNET**, *Student Internet Service Provider*, Évry (91), France.
- Jun 2019 Volunteer in the free-software student association. Internet access provider and network infrastructure for the on-campus housing. Approx. 850 rooms in 7 buildings, within a team of 15 admins.
- **IT support**: help-desk activities and free-software promotion to all students on campus.
 - **Event organization**: Linux install party, free software & technical annual conference, massive triennial student night party with more than 1000 attendees.
 - **IT Admin**: (*SYS*) Debian, Proxmox VE, (*NET*) Cisco, (*Software Dev*) Ruby, Python, Bash.
- May 2016 **Training supervisor at MiNET**, *Student Internet Service Provider*, Évry (91), France.
- May 2017 In charge of technical trainings for new members (student administrators).

Publications

Conference

- MICRO'26** *Synapse: Decoupling Protocol, Topology, and Policy for Automated Synthesis of Coherence Controllers*
Nicolò Carpentieri, **Anatole Lefort**, David Schall, Julian Pritzi, Pramod Bhatotia.
In *59th IEEE/ACM International Symposium on Microarchitecture (MICRO)*, Athens, Greece, 2026
- ASPLOS'26** *vCXLGen: Automated Synthesis and Verification of CXL Bridges for Heterogeneous Architectures*
Best paper award **Anatole Lefort**, Julian Pritzi, Nicolò Carpentieri, David Schall, Simon Dittrich, Soham Chakraborty, Nicolai Oswald, Pramod Bhatotia.
In *Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, Pittsburgh (PA), USA, 2026
- HPCA'26** *C³: CXL Coherence Controllers for Heterogeneous Architectures*
Anatole Lefort, David Schall, Nicolò Carpentieri, Julian Pritzi, Nicolai Oswald, Pramod Bhatotia.
In *Symposium on High-Performance Computer Architecture (HPCA)*, Sydney, Australia, 2026
- SOSP'21** *J-NVM: Off-heap Persistent Objects in Java*
Anatole Lefort, Yohan Pipereau, Kwabena Amponsem, Pierre Sutra, Gaël Thomas.
In *Symposium on Operating Systems Principles (SOSP)*, Virtual, 2021
- DSN'19** *White-Box Atomic Multicast*
Alexey Gotsman, **Anatole Lefort**, Gregory V. Chockler.
In *Dependable Systems and Networks (DSN)*, Portland (OR), USA, 2019

Journals

- TOCS** *CXL Bridges for Multi-Host Coherent Memory across Heterogeneous Architectures*
Invited extended journal version through recommendation by the PC chairs of ASPLOS '26
Anatole Lefort, Julian Pritzi, Nicolò Carpentieri, David Schall, Simon Dittrich, Soham Chakraborty, Nicolai Oswald, Pramod Bhatotia.
In *ACM Transactions on Computer Systems (TOCS)*, forthcoming, 2026.

Talks

- Conference Talk** In *Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, Pittsburgh (PA), USA, 2026, *vCXLGen: Automated Synthesis and Verification of CXL Bridges for Heterogeneous Architectures*
- Conference Talk** In *Symposium on High-Performance Computer Architecture (HPCA)*, Sydney, Australia, 2026
C³: CXL Coherence Controllers for Heterogeneous Architectures
- Conference Talk** In *Symposium on Operating Systems Principles (SOSP)*, Virtual, 2021
J-NVM: Off-heap Persistent Objects in Java
- Workshop Paper** In *Non-Volatile Memories Workshop (NVMW)*, San Diego (CA), USA, May 2022
J-NVM: Off-heap Persistent Objects in Java.
- Invited Talk** In *Heterogeneous Memory Workshop (HMEM)*, Virtual, July 2022
J-NVM: Off-heap Persistent Objects in Java

Professional Service

- Program Committee member **EuroSys'27**
- Program Committee member **SOSP'26** Posters track
- Program Committee member **HCDS'26** (*Workshop on Heterogeneous Composable and Disaggregated Systems*)
- Artifact Evaluation Committee of **SOSP'23**

Prizes and Awards

- ASPLOS'26 Best paper award**
for the paper *vCXLGen: Automated Synthesis and Verification of CXL Bridges for Heterogeneous Architectures*.
- Laureate of “Engineers of the Future Awards”, Research Engineer category, 2022 Edition.**
Sponsored & Issued by *L'Usine Nouvelle* - a renown french business magazine.
- Best student publication in ICTs at Institut Polytechnique de Paris, 2022 Edition.**
Sponsored & Issued by *Labex DigiCosme*, *Institut Polytechnique de Paris* and *Université Paris-Saclay*.

Grants

- **Humboldt Research Fellowship for Postdocs** from *the Alexander von Humboldt Foundation*, 2023-2025.
- **NVMW Student Travel Grant** to attend *NVMW'22* (San Diego, CA, USA).
- **Fully funded Ph.D. scholarship** from *Institut Mines-Télécom*, “*Future & Rupture*” campaign (2018). Awarded on *Academic Excellence* criteria, ranked 1st for *Télécom SudParis*.

Teaching Experience

- **Practical Lab: System Programming (C, Rust):**
Co-coordinator Winter 2024-25, Summer 2025
Teaching Assistant Winter 2023-24, Winter 2025-26, Winter 2026-27
- **Lecture: Introduction to Software Engineering (Java):**
Lecturer & Exam writing Summer 2024, Summer 2026
- **CSC 3102: Operating systems, Unix & Shell scripting (Bash):**
Lecturer Fall 2020
Teaching Assistant Fall 2019
- **CSC 3101: Data structures & algorithms (Java 1):**
Teaching Assistant Fall 2020
- **CSC 4102: Object-oriented programming & software engineering (Java 2):**
Teaching Assistant Spring 2020, Spring 2021

Student Supervision

- Nicolò Carpentieri, PhD, Fall 2024 - ongoing, *Coherence Controllers for Heterogeneous Architectures*
- Nguyen Le, Bsc, Summer 2026, *Heterogeneous Litmus Tests for CPU-GPU Memory Consistency*
- Ivan Logvynenko, Bsc, Summer 2026, *Flat Memory for CXL 3.0 GPUs*
- Victor Trost, Bsc, Winter 2025, *Disaggregated and Vendor-Generic APUs with CXL*
- Julian Pritzi, Msc, Winter 2024, *Formal Verification of Heterogeneous Cache Coherence Protocols for CXL*
- Simon Dittrich, Msc, Winter 2024, *Generating Heterogeneous Cache Coherence Protocols for gem5 Simulations*
- Mouhamadou Mangassouba, Msc, Spring 2022, *P-FaaS: Stateful Serverless with Persistent Memory*
- Kwabena Amponsem, Msc, Fall 2020, *P-Transformer: A Java Bytecode Transformer for Persistent Objects*

Skills

Computing

Programming Java, C, Python, Bash, SQL, Latex, Go, OCaml, B-method, PHP, Ruby

Systems Linux, Unix, Cisco IOS

IT Admin

Virt. Proxmox VE, QEMU/KVM, LXC, Docker

Services FreeRadius, Apache, Nginx, OpenVPN, Zabbix, ISC-KEA, ELK

Languages

English European C1 level - TOEIC 970

French Native speaker

German European A2 level

Spanish Beginner

Other Interests

Hobbies Computer tinkering (hardware and software); problem solving.
Freeline skating, Outdoor activities.

Sports Cycling (Road, MTB, Touring, Bikepacking),
Running (Road or Trail), Swimming
Hiking & Backpacking, Skiing, Snowboarding

Past Sports Gymnastics, Tennis, Judo